P07 | Contribution, evidence and closure

For David Celeita and Masoud Barati | Coauthor-review draft | 31 Oct 2026, 17:12 Colombia

Question and contribution

Can eight PMUs distinguish a new load/line intervention from the electrical recovery of an earlier one? P07 specifies a causal physical-history attribution procedure: retained states survive restoration, new interventions compete with recovery, and active-event mass is distinguished from originating-cause mass. This connects the physical cause, its active interval and the remaining response in one sequential explanation, tested against operational errors rather than accuracy alone.

Where physics and Bayes enter

Kirchhoff constraints, Ybus, oriented branch currents and machine/control dynamics determine reduced DAE predictors. Known PMU locations select the observation map; phase and instrument memory supply frequency/ROCOF predictions. States and operating context come from past PMU reports. Conditional Gaussian evidence integrates truncated load severity; predictive likelihoods weight five constant R/X hypotheses. Onset screening, shared regime covariances, pruning and publication rules make the complete procedure approximate.

Completed evidence: development, not C01 confirmation

H04 procedure	Timely family + source	Extra alarms
Physical: five models	53/54	4
Physical: nominal	54/54	7
Both frozen ExtraTrees adapters	27/54	3

54 realizations repeat 18 interventions across three network conditions; normal exposure is only 204 s per method. Nominal/mixture publication paths also differ, so this is not an isolated R/X-bank ablation.

What the controlled experiments establish

With load family and ON/OFF supplied, recentering and conditioned signatures both recover 240/240 sources (80 physical event cases, three shared-noise realizations). Conditioning lowers median relative response error 1.25% to 0.0561% and severity error at the generating source 1.2% to 0.329%. A separate 38-trajectory recovery control finds no mean binary-localization gain (98.35% for both arms; 12 correlated, known-history/onset comparisons). Better fit, source identity and calibrated uncertainty are distinct outcomes.

Difference from the hackathon and the literature

The contribution changes the explanation and temporal evaluation, rather than adding a source-label classifier. H04 favors the physical procedures on loads but not on extra-alarm count; it is not an official leaderboard comparison. Warner already uses physical trajectories and onset search; IMM/M3H already retains mode histories. No numerical advantage over an unexecuted matched M3H/Warner method is claimed.

Limits and the fixed route to submission

C01: 213/342 records complete at this snapshot; 171 physical trajectories have finished. Estimated remaining processing is 25 h (recent timing envelope 21-33 h), plus evaluation/build. Finish the frozen comparison of R11, nominal and both ExtraTrees adapters; score Chen2016 on persistent openings. Audit all intended cases, normal exposure, delays, censoring and parent-paired intervals, then finalize claims with the coauthors. Scope remains nine load and nine line assets, an event-free startup and a declared simulator. No field, unseen-source, all-family, real-time or calibrated-probability claim; BUS5 terminal and BUS39 angle limitations remain. No new method versions or test tuning.